Strong lensing modeling with deep learning

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Abstract

Gravitational lensing is a powerful probe of the dark matter distribution in galaxies and an independent means of constraining the Hubble constant, (H_0). These applications require accurate inference of lens mass distributions from observed lens images. While traditional approaches based on Markov Chain Monte Carlo (MCMC) sampling are computationally intensive, deep-learning methods enable parameter estimation in milliseconds. We model lensing galaxies as Singular Isothermal Ellipsoids (SIEs) with external shear and investigate several deep-learning architectures trained on simulated lens images representative of observations from the Hyper Suprimer-Cam (HSC) survey on the Subaru Telescope. The networks are designed to infer five lens parameters: the Einstein radius, the ellipticity components e_x and e_y, and the external shear components $\gamma_{\text{ext, 1}}$ and $\gamma_{\text{ext, 2}}$. Staring from a standard Residual network (ResNet), we incorporate a U-Net segmentation model to enhance the extraction of lensing features such as arcs and Einstein rings. We further extend the architecture to a Bayesian Neural Network that predicts both parameter estimates and their associated 1σ uncertainties. The Einstein radius is recovered with high accuracy with 68% of the predictions showing relative errors between -3% and +3%, and the ellipticity components are inferred robustly. In contrast, the external shear parameters remain more challenging to constrain, highlighting the need for further improvements in model architecture and training strategies. The codes and models can be found at the following GitHub : `https://github.com/matthbatt/Deep-learning-gravitational-lensing`

1 Introduction

History

In 1915, Einstein published his work on general relativity, which fundamentally changed our understanding of the Universe [1]. Gravity is now seen as a curvature of spacetime rather than a force. A direct consequence of this framework is *gravitational lensing*, a phenomenon in which light is deflected by a gravitational field.

During the Solar eclipse of 29 May 1919, British astronomer Arthur Eddington conducted an observational experiment to test the prediction of general relativity. Two expeditions were deployed : One in Brazil the other one on the west coast of Africa, where astronomers photographed stars during the total eclipse when the moon blocked the sun. The experiment revealed that the stars' positions were displaced by the sun's gravitational field, confirming Einstein's theory of general relativity and making him instantly famous.

This is a case of *weak gravitational lensing*, in which the gravitational field of the lens is just strong enough to distort the shapes of objects behind it. In 1936, Einstein demonstrated that a sufficiently strong gravitational field could cause a lens to produce multiple images and arcs of a background object. However, this case of *strong gravitational lensing* was initially thought to be unobservable, as the phenomenon was considered too rare and too small to detect with the technology available at that time.

It was not until 1979 that the first strong lensing system was observed, the so-called Twin Quasar (see the Wikipedia page : *Twin Quasar Q0957+561*), a quasar whose light has been distorted by a massive galaxy in the foreground, creating two images of the same source. With their insanely strong gravitational field, galaxies and clusters of galaxies are excellent candidates for strong lensing to occur (see the ESA/Hubble page for some impressive gravitational lensed images : *ESA/Hubble lenses*.

Motivation for strong lensing

In addition to being a visually striking phenomenon, strong lensing is also a powerful tool for studying the universe. In particular, measuring the total mass of the lens enables the study of the mass structure of galaxies, and thereby estimating their dark matter distribution [2]. Moreover, lensing magnifies the source's light, allowing the study of very distant objects [3]. If the source light's is varying, such as in a quasar or a supernova, measuring the time delays between the appearance of the multiple lensed images can provide an independent probe of the Hubble constant H_0 [4], which quantifies the expansion rate of the universe.

Dark matter distribution

To estimate the dark matter distribution, one can consider the example of the analysis performed on the Cosmic Horseshoe strong gravitational lensing system. Schuldt et al. [2] inferred the dark matter distribution by combining strong lensing observations from the Hubble Space Telescope with stellar dynamic measurements. They modeled the galaxy as a combina-

tion of visible (stellar) matter and dark matter, testing different dark matter density profiles to determine which mass distribution best produces the observed Einstein ring, radial arc, and stellar velocity dispersion. Among these profiles, they considered the Navarro-Frenk-White (NFW) profile, which describes how the dark matter density changes as a function of distance from the galaxy center. They also included the velocities of stars through dynamical modeling, providing additional constraints on the galaxy's gravitational potential and allowing them to better separate the contributions of luminous matter and dark matter.

Distant object study

In 2018, a cosmological team conducted a detailed study of the distant galaxy J1414+5446, located at a redshift of (z=5.4). This galaxy is one of the brightest known galaxies at redshift $z > 5$ [3]. Its exceptional luminosity provides a unique opportunity to investigate the physical properties of an individual galaxy near the epoch of reionization. Moreover, it offers a preview of the type of observations that will become possible with upcoming wide-area surveys, which are expected to discover hundreds of similarly bright high-redshift galaxies.

The Hubble constant

Currently, H_0 can be estimated using standard candles such as Cepheid variables or Type Ia supernova. Because their intrinsic luminosity is known, their distance can be inferred from their observed brightness, allowing H_0 to be determined from the relation between distance and redshift [5].

Another technique relies on observations of the *Cosmic Microwave Background (CMB)*, the relic radiation emitted approximately 300,000 years after the Big Bang. By fitting a cosmological model to this radiation, measurements of the CMB can be used to infer H_0 [6].

However, a significant discrepancy currently exists between the values of H_0 inferred by the CMB and those obtained by the local distance measurements. This so called *Hubble Tension* has become one of the most important open questions of modern cosmology.

Strong gravitational lensing provides an independent method for measuring H_0, which may help to resolve this tension. In time-delay cosmography, H_0 is inversely proportional to the time delay between the arrival of light from multiple lens images of the same source. Therefore, by accurately measuring these time delays and modeling the lens system, it is possible to infer a value of H_0.

Motivation for deep learning

For decades, the number of known and studied gravitational lenses remained small, as this phenomenon requires an almost perfect alignment between a massive object, the observer, and the background source. In fact, some studies have shown that fewer than one out of every 10^{3-4} galaxies exhibits lensing features [7].

However, this situation is expected to change. With new-generations surveys such as the ESA *Euclid* satellite [8] and the *Vera Rubin Observatory* [9], more than a billion galaxies will be observed, enabling the identification of up to 10^5 gravitational lenses. To analyze these large datasets, significant efforts have been made to develop machine learning techniques for the rapid identification of lens candidates[10].

Theses machine learning techniques can also be used to model detected lenses, as traditional methods such as Bayesian analysis, Monte Carlo Markov Chain (MCMC) and nested sampling are computationally expensive and require non-trivial human intervention for lens selection [11], making them unsuitable for modeling large numbers of gravitational lenses. Several studies have applied machine learning algorithms [15] [14], like Convolutional Neural Network (CNN) [12], which can model large numbers of gravitational lenses in milliseconds for a pre-trained network.
In this work, we aim to implement different machine learning architectures to model detected strong lenses and evaluate their performance. Section 2 presents the theoretical framework of gravitational lensing, with particular emphasis on the lens parameters to be inferred. In Section 3, we introduce the deep-learning methods considered in this study, including the ResNet architecture and the simulated dataset used for training and evaluation. We then assess the performance of the ResNet model and subsequently investigate the impact of incorporating a U-Net architecture. We also introduce a Bayesian Neural Network that provides both parameters estimates and associated uncertainties. Finally we tested the architectures on datasets simulated from Euclid high resolution images. All models were trained using one GPU and three CPUs.

2 Lensing theory

To study dark matter distribution and constrain the Hubble constant, one must correctly model the lens galaxy (Fig. 1) using the lens equation :

$$\beta = \theta - \alpha$$

where β is the source position in the source plane, θ the image position in the image plane and α is the deflection angle, (see Fig. 1).

The deflection angle α depends on the lensing potential : $\alpha = \nabla\psi_{\rm SIE} + \nabla\psi_{\rm Ext}$. In this study, we model the lensing galaxies using a Singular Isothermal

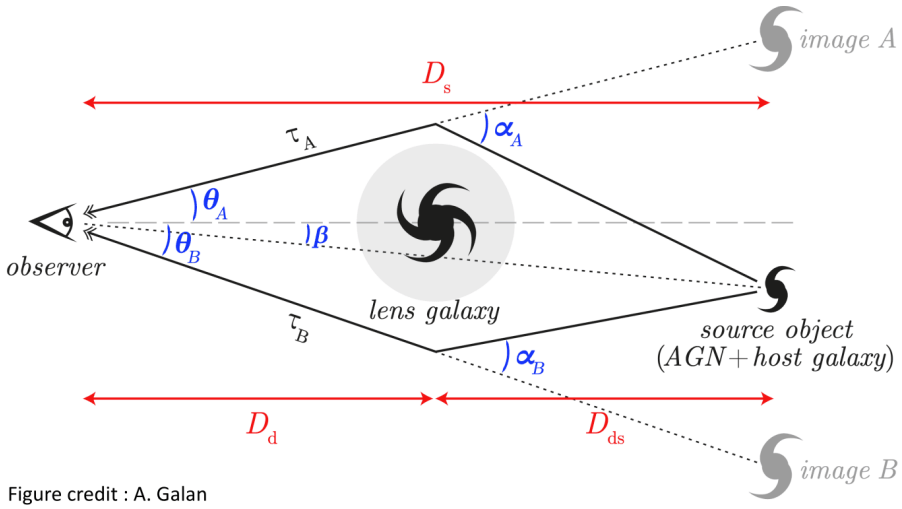


Figure 1: Strong lensing diagram

Ellipsoid (SIE) model and include external shear. For most massive and isolated galaxies, it is appropriate to use the SIE model, which assumes that galaxy is elliptical and that the stellar density decreases as r^{-2} with increasing distance from the center.
The external shear quantifies the additional deformation induced by surrounding galaxies on the lensing system.

Lensing potential parameters

In this work, using the SIE potential ψ_{SIE} requires the following parameters :

- **Einstein radius θ_E :** The angular radius of the ring-shaped image.
  - ❖ $\kappa(r) = \frac{\theta_E}{(1+q)r}$

  which depends on the convergence κ or dimensionless surface mass density, elliptical radius r and axis ratio q :
  - ❖ $q = \frac{\sqrt{1-\sqrt{e_x^2+e_y^2}}}{\sqrt{1+\sqrt{e_x^2+e_y^2}}}$
- **Complex ellipticity e_x and e_y :** These are the complex ellipticity that describes the shape of the lensing galaxy.
  - ❖ $e_x = \frac{1-q^2}{1+q^2}\cos(2\phi)$
  - ❖ $e_y = \frac{1-q^2}{1+q^2}\sin(2\phi)$

  where ϕ is the orientation angle ellipse's major axis.

For the external shear potential ψ_{ext} we model with the following parameters :

- **Complex external shear $\gamma_{\text{ext, 1}}$ and $\gamma_{\text{ext, 2}}$:**
  - ❖ $\gamma_{\text{ext, 1}} = \gamma_{\text{ext}}\ \cos(2\phi)$
  - ❖ $\gamma_{\text{ext, 2}} = \gamma_{\text{ext}}\ \sin(2\phi)$

  with ϕ the orientation angle of the shear field, and γ_{ext} being the total shear strength :
  - ❖ $\gamma_{\text{ext}} = \sqrt{\gamma_{\text{ext, 1}}^2 + \gamma_{\text{ext, 2}}^2}$

In total, we consider five parameters : $\theta_E, e_x, e_y, \gamma_{\text{ext},1}, \gamma_{\text{ext},2}$. Our goal is to build a deep learning model capable of inferring these parameters from an image of strong gravitational lensing system. Before describing our approach, we first introduce the concept of a deep learning model.

3 Deep Learning models

Basic Introduction to Neural Network

Neural networks are computational models inspired by the structure of the human brain. A neural network is composed of interconnected processing units called *neurons*. Each neuron receives inputs from other neurons, applies a mathematical transformation, and passes its output to subsequent neurons. A neuron is characterized by a set of weights and an activation function.

For example, imagine a neuron receiving outputs from three other neurons. Let these outputs be x_1, x_2 and x_3 (input of our neuron) and let the corresponding connection weights be a_1, a_2 and a_3. The neuron computes a weighted sum of its inputs and adds a bias term b_0. This result is then passed through an activation function f.

The output of the neuron is given by:

$$y = f(a_1x_1 + a_2x_2 + a_3x_3 + b_0)$$

Where:
(x_1, x_2, x_3) are the inputs to the neuron, (a_1, a_2, a_3) are the weights associated with each input, (b_0) is the bias (or intercept), (f) is the activation function, y is the output of the neuron.

Common activation functions include the Rectified Linear Unit (ReLU), defined as $(\max(0, x))$, as well as

sigmoid, hyperbolic tangent (tanh), and arc-tangent functions. The activation function introduces non-linearity into the network, allowing it to learn complex relationships within the data.

How Do Neural Network Learn?

Many AI algorithms can be viewed as optimization problems. In the case of neural networks, the objective function to be minimized is the *Loss Function*, which quantifies the difference between the model predictions and the target values. The optimization is performed over the network parameters, namely its weights and biases. The learning process occurs through two main steps:

- **Forward propagation :** the input data passes through the network, producing a prediction ($\hat{y}$).
- **Backward propagation :** the prediction is compared to the true value (y) by computing the *loss function* : $\mathcal{L}$, and the network's parameters are adjusted to reduce this loss. The technique used is *gradient descent*, which consists of updating the parameters of each neuron according to the gradient of the loss function $\mathcal{L} : a \rightarrow a - lr \frac{\partial \mathcal{L}}{\partial a}$ with lr a hyperparameter known as the *Learning rate*.

Neural networks are organized into layers. Each neuron in one layer is typically connected to several neurons in the next layer. If a layer contains (N) neurons and the next layer contains (M) neurons, there are generally ($N \times M$) connections between them. As the number of layers increases, the network becomes deeper. Increasing the depth of the network is often more effective than increasing its width because deeper networks can learn hierarchical feature representation and express complex functions with significantly fewer parameters than shallow networks.

However, increasing the depth of a neural network can make the learning process more challenging, as information may gradually degrade due to vanishing gradient while propagating thought successive layers. Let us see why the *ResNet* model can solve this issue.

3.1 ResNet model

Residual neuron network (ResNet) are deep learning architectures where each block learns a residual function with reference to its input, instead of learning the full function from scratch. In standard machine learning, the model learns the mapping (H) between the output (y) and the input (x) :

$$y = H(x)$$

But when the networks become deep, the mapping (H) can be difficult to learn. So, the philosophy of a ResNet is to learn the residual F instead of the full mapping :

$$F(x) = H(x) - x$$

Thus, the ResNet learns $y = H(x) + x$, which requires learning small adjustments to go from x to y, instead of learning the whole function. In terms of architecture, a residual block is just a set of layers with a skip connection (see Fig. 2).

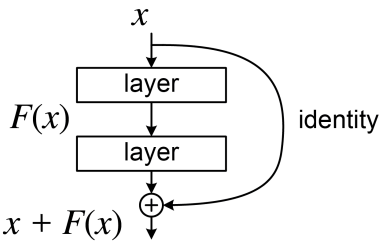


Figure 2: ResNet residual block

This architecture, developed in 2015, made huge improvements in deep learning. It allowed networks to reach extreme depths (50, 101, 152 layers) without the vanishing gradient.

Our ResNet architecture

In this work, we use a ResNet architecture as our benchmark model. The implemented network has approximately 1.26 million trainable parameters and it is composed of four residual blocks (see Fig. 3).

As input, the ResNet receives four images of size 224 by 224. The first residual block produces 32 feature maps of size 224 by 224, while successive blocks progressively reduce the spacial resolution, reaching 256 feature maps of size 14 by 14 in the final residual block. Global average pooling is applied by taking the mean value of each the 256 feature maps, producing a 256 dimensional feature vector. This feature vector is passed to a regression head consisting of a fully connected layer of 256 by 128 by 5 connections (see Fig. 3) which predicts the five target parameters.

Now we introduce the dataset used to train the ResNet model.

The Dataset

The dataset consists of 43,733 simulated strong gravitational lenses. The background sources are taken from Hubble Ultra Deep Field images, corresponding to a small region of the sky in the Constellation Fornaxr. The foreground lensing galaxies are photographed by the Subaru telescope, an 8.2 meter instrument located near the summit of Maunakea, Hawaii, operated by the National Astronomical Observatory of Japan. A lensing transformation is applied to the background light, as if it had passed through the gravitational potential of the foreground galaxy, producing arcs and rings. Noise is then added to the result, simulating observations made with the Subaru telescope.

Each simulation has two extensions : One containing the lensed background source together with the foreground lensing galaxy, and a second containing the

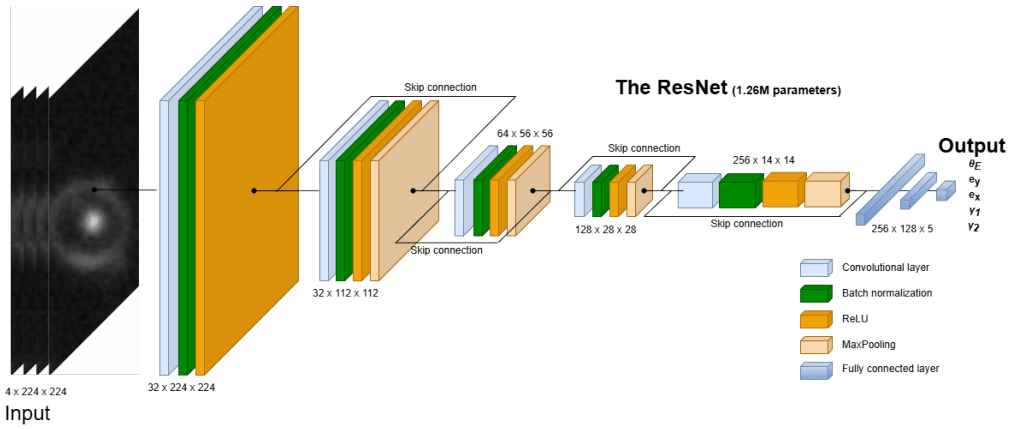


Figure 3: Our ResNet model

lensed background source only. Both configurations include four bands : g–(400-550nm), r–(550-700nm), i–(700-820np) and z–(820-920nm), making a total of eight extensions for each simulation.

Figure 4 presents six examples of the simulated lensing systems. In each image, the bright object at the center is the lensing galaxy, taken from Subaru observations, while the surrounding arcs and rings correspond to background source galaxies extracted from the Hubble Ultra Deep field.

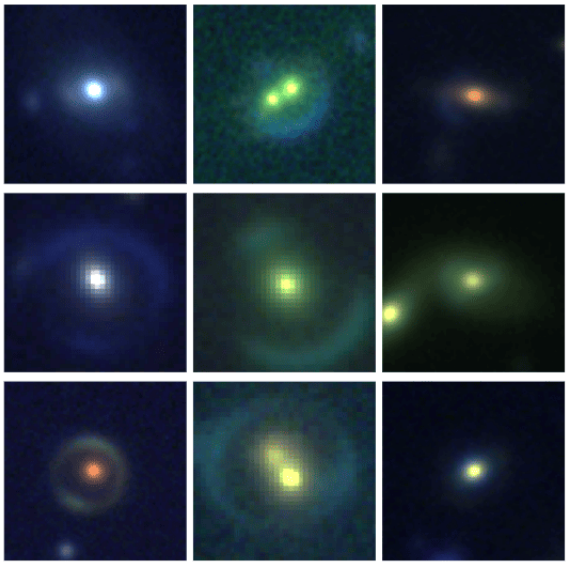


Figure 4: Six examples of the colored mock lenses system from the dataset, produced by combining the g, r and i filters.

Parameters to estimate

In this work, We consider 5 parameters to estimate

- **The Einstein radius θ_E :** Values are uniformly distributed (see Fig. 5) between 0.5 and 2.5 arcseconds, allowing the model to learn across the full range of the parameter space as suggested in [16].
- **Ellipticity e_x and e_y :** Values lie between -0.7 and +0.7 arcseconds, with higher concentration of samples around 0.
- **External shear components $\gamma_{\mathbf{ext,1}}$ and $\gamma_{\mathbf{ext,2}}$:** They account for the influence of nearby structures in the image plane. In the simulations, these parameters are not derived from actual surrounding galaxies but are instead added as random perturbations to the mass profile. Their values range from -0.1 and $+0.1$ arcseconds, with most values concentrated around 0.

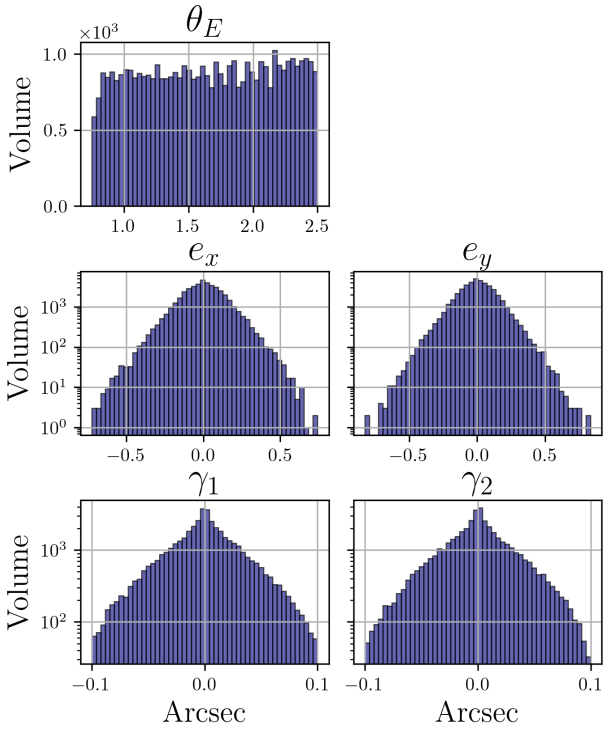


Figure 5: Histogram of the five parameters to estimate.

ResNet Training

To train our ResNet model we split the dataset into 60% training, 20% validation and 20% testing sets.

The number of epochs is fixed to 100, meaning the model processes the entire training set 100 times. However, the full dataset is not processed at once. Instead, it is divided into smaller groups called *batches*. The batch size is set to 32, indicating the model operates on 32 samples simultaneously. For each batch, the model computes the loss function which is the Mean Squared Error (MSE) :

$$\mathcal{L}oss = \frac{1}{32} \sum_{i=1}^{\text{Batch: } 32} (\hat{\mathbf{Y}}_\mathbf{i} - \mathbf{Y}_\mathbf{i})^2 \tag{1}$$

where $\mathbf{Y_i}$ represents the ground-truth five parameter vector of sample i $(\theta_E, e_x, e_y, \gamma_{\text{ext},1}, \gamma_{\text{ext},2})$ and $\hat{\mathbf{Y}}_\mathbf{i}$ is the corresponding the five parameter prediction produced by the model.

The model then updates its weights using backward propagation. Once all batches in the training set have been processed, the model is evaluated on the validation set by computing the loss function without updating any weights. This validation step is very useful to monitor the model's performance and to detect overfitting. Overfitting occurs when the model relies too heavily on the training data and is not capable to generalize its predictions to new dataset. After both sets are completed, the model proceeds to the next epoch.

We apply early stopping, which halts training when the validation loss isn't improving for 20 consecutive steps. We also save the model parameters corresponding to the lowest validation loss during training.

For optimization, we use the Adam optimizer with a weight decay of 10^{-4} to reduce overfitting. The initial learning rate is set to 10^{-3}, and it is adjusted using PyTorch's CosineAnnealing schedule, which gradually reduces the learning rate following a cosine curve for smoother convergence.

We also apply several transformations to the input images. In the simulations, the lensing galaxy is always located at the center of the image. To prevent the model from expecting the lens to always be located at the center of the image, we randomly shifts the image center so the lensing galaxy appears at different locations. we also evaluated different intensity transformations including the hyperbolic arcsin transformation, the square-root scaling and image normalization. The best performances was obtained by first setting the all negative pixels to 0 and then applying a square-root transformation.

ResNet's Results

In this section we present the performance of the ResNet on the test set containing approximately 8,000 samples. Fig. 6 shows the evolution of training and validation losses throughout the training process. The training is stable: the validation loss reaches a plateau after 10 epochs, while the training loss continues to decrease. Early stopping is triggered at epoch 40, with no increase of the validation loss, indicating no evidence of overfitting. The total training time of the ResNet is approximately one hour.

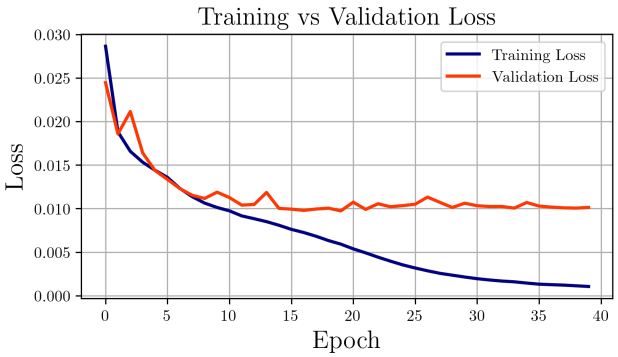


Figure 6: ResNet's loss function during training

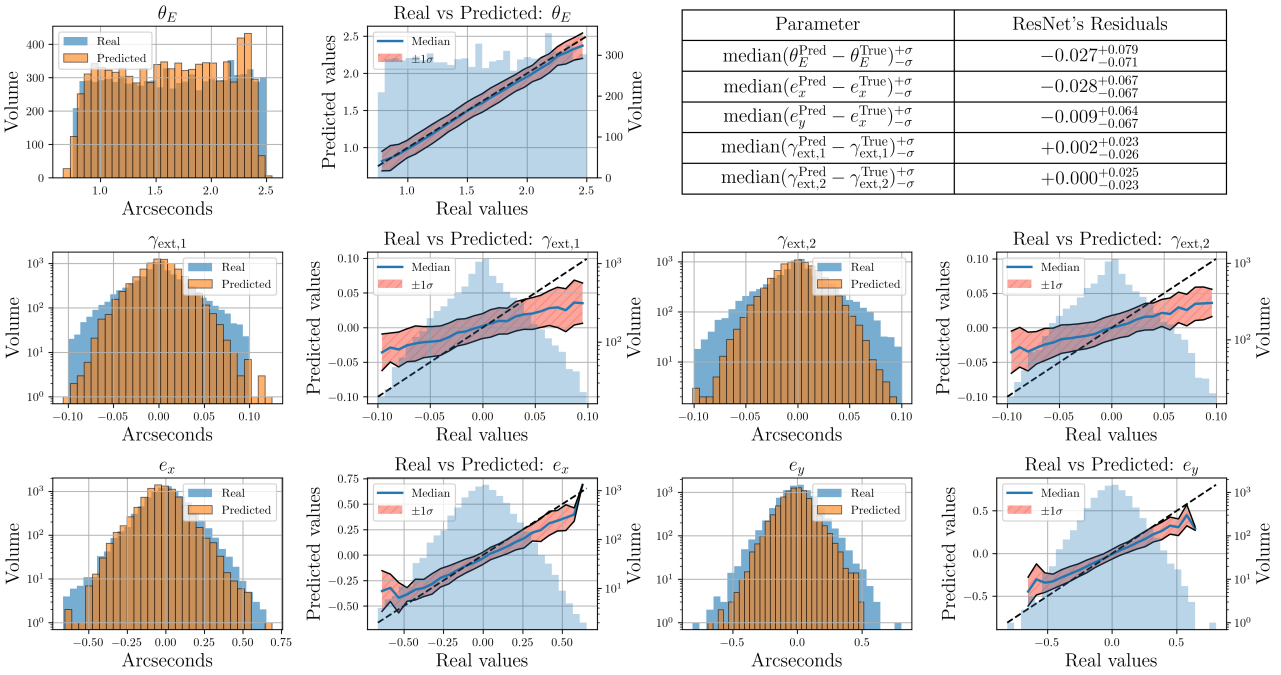
Discussion about the predictions (Fig. 7) :

Einstein Radius θ_E : This parameter is the best predicted among the five. The distribution of the predicted values closely matches that of the ground truth (left plot). Furthermore, the median line closely follows the diagonal line (right plot), while the 1 σ confidence band remains narrow and nearly constant across the entire parameter range, indicating homoscedastic residuals. The residuals statistics reported in the table confirm this excellent performance. The median residual is small with a bias of only -0.027 arcsec, which corresponds to an error of 5%. The upper and lower 1σ intervals are slightly asymmetric (0.08 and 0.07 arsec, respectively) indicating a mild skewness in the residual distribution.

Complex ellipticity e_x and e_y : At first glance the model appears to recover the ellipticity to a reasonable extent. The distribution plots indicate that the predicted distribution of e_x matches the ground-truth's distribution more closely than that of e_y. For e_x, noticeable discrepancies appear only for values below -0.5 and above 0.5. In contrast, the agreement for e_y deteriorates more, rapidly, particularly for values larger than 0.5 where the network predicts virtually no samples. The deviations become apparent at smaller absolute values than for e_x. On the second plot, both median curves move away from the diagonal line, indicating a reduced ability of the model to reproduce the tails of the ellipticity distributions.

External shear $\gamma_{\text{ext, 1}}$ and $\gamma_{\text{ext, 2}}$: These are the two parameters for which the model struggles the most to predict. The discrepancy between the true and the predicted distributions appears rapidly as soon as we move away from the mean value of 0. There are three possible reasons for this difficulty :

1. **Limited resolution of ground-based images:** The simulated images are degraded to match observations from the Subaru telescope. Since shear corresponds to a small deformation added to the lensing system, its signal can easily be absorbed by the image's noise.



Parameter	ResNet's Residuals
$\text{median}(\theta_E^{\text{Pred}} - \theta_E^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.027^{+0.079}_{-0.071}$
$\text{median}(e_x^{\text{Pred}} - e_x^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.028^{+0.067}_{-0.067}$
$\text{median}(e_y^{\text{Pred}} - e_x^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.009^{+0.064}_{-0.067}$
$\text{median}(\gamma_{\text{ext},1}^{\text{Pred}} - \gamma_{\text{ext},1}^{\text{True}})^{+\sigma}_{-\sigma}$	$+0.002^{+0.023}_{-0.026}$
$\text{median}(\gamma_{\text{ext},2}^{\text{Pred}} - \gamma_{\text{ext},2}^{\text{True}})^{+\sigma}_{-\sigma}$	$+0.000^{+0.025}_{-0.023}$

Figure 7: Performance of the ResNet on the test set (approx. 8000 samples). For each of the five parameters, two panels are shown. The left panel compares the distribution of the predicted values the ground truth. The right panel shows the prediction values as function of the ground-truth values. The median prediction (solid blue line) and the corresponding 1 σ confidence band are overlaid. The table in the upper-right corner summarizes the median residuals and their corresponding 1 σ intervals for all five parameters.

2. **Imbalanced dataset distribution :** The Dataset is strongly concentrated around 0, which biases the model toward predicting values close to the mean. This effect is visible in the median curve, which appears flatter than the diagonal line.

3. **Ellipticity and external shear degeneracy :** Ellipticity and external shear have similar effects on the lensing potential. Different combinations of ellipticity and external shear can therefore produce indistinguishable lensed images. As a result, instead of predicting a strong external shear, the model may compensate by predicting a higher ellipticity. Figure 8 reveals a weak but noticeable orientation of the 2D density distribution of Δe_x versus $\Delta\gamma_{\text{ext, 1}}$ and Δe_y versus $\Delta\gamma_{\text{ext, 2}}$, suggesting a correlation between ellipticity and external shear.

Overall, the predictions remain highly accurate. Comparing our results with those from [16], we find a notable improved performance for the Einstein radius. This may be attributed to the simulated arcs being brighter and to the more uniform distribution of Einstein radius in the dataset. In the next section we will present a strategy to improve prediction performance.

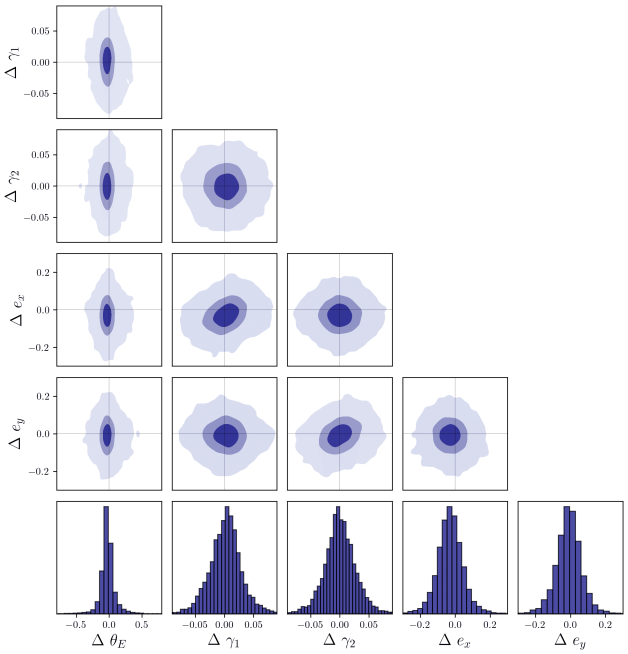


Figure 8: 2D density plots of the residuals for the five parameters, together with their corresponding marginal distribution in the bottom row of the figure.

3.2 U-Net model

A *U-Net* model is a deep learning architecture with a U-shaped design that is highly effective for image segmentation tasks. It consists of an encoder part,

that progressively compresses image information into a lower-dimensional latent representation (the bottleneck), and a decoder that reconstructs the spacial resolution to produce a segmentation map. Similarly to ResNet architectures, U-Net also incorporates skip connections between corresponding encoder and decoder layers, which helps to preserve fine details of the image during reconstruction.

In this work, the U-Net is used to remove the nearby galaxies surrounding the lensing system and the lensing galaxy allowing the subsequent ResNet to focus exclusively on the arcs and rings themselves for parameter prediction (see Fig. 9).

To train the U-Net + ResNet model, the U-Net must first learn to isolate the lens component from the input images. In the mock lens dataset, each sample includes a simulated lensing system observed in four bands, five parameters to be estimated, and the corresponding ground-truth lens-only image used to generate the full simulation. The U-Net is trained by comparing its output to the ground-truth lens image, resulting in a reconstruction loss. Consequently, the overall training procedure involves two loss functions :

$$\mathcal{L}oss = \mathcal{L}oss_{\text{U-Net}} + \beta\ \mathcal{L}oss_{\text{ResNet}} \tag{2}$$

where $\mathcal{L}oss_{ResNet}$ is defined in eq. 1 and $\mathcal{L}oss_{\text{U-Net}}$ is the reconstruction loss by the U-Net. It is defined as the Mean Squared Error between the U-Net's extracted lens and the ground-truth lens only image. β is a hyperparameter used to weight the 2 losses.

The U-Net and the ResNet are trained jointly, with the parameter β fixed at 0.01 to ensure that both loss terms are of comparable magnitude. The learning rate is initialized at 10^{-3} and is gradually decreased using the cosine annealing scheduler. The weight decay is fixed at 10^{-4}. The training time is now approximately 8 hours, with early stopping typically triggered at epoch 80. The learning is stable with no traces of overfitting (see Fig. 10).

To access the prediction accuracy of the U-Net + ResNet architecture we compare its residuals with the ResNet in table 1.

Parameter	**ResNet**	**UNetResNet**
$\text{med}(\theta_E^{\text{Pred}} - \theta_E^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.027^{+0.079}_{-0.071}$	$-0.002^{+0.049}_{-0.048}$
$\text{med}(e_x^{\text{Pred}} - e_x^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.028^{+0.067}_{-0.067}$	$-0.002^{+0.067}_{-0.066}$
$\text{med}(e_y^{\text{Pred}} - e_y^{\text{True}})^{+\sigma}_{-\sigma}$	$-0.009^{+0.064}_{-0.067}$	$+0.003^{+0.071}_{-0.067}$
$\text{med}(\gamma_1^{\text{Pred}} - \gamma_1^{\text{True}})^{+\sigma}_{-\sigma}$	$+0.002^{+0.023}_{-0.026}$	$+0.000^{+0.022}_{-0.027}$
$\text{med}(\gamma_2^{\text{Pred}} - \gamma_2^{\text{True}})^{+\sigma}_{-\sigma}$	$+0.000^{+0.025}_{-0.023}$	$+0.001^{+0.022}_{-0.022}$

Table 1: Comparison of prediction residual statistics between ResNet and U-Net + ResNet models.

The residual statistics indicate a significant improvement in the bias for the Einstein radius θ_E and the ellipticity component e_x, with the bias reduced approximately by an order of magnitude. The Einstein radius predictions achieve high accuracy, with 68% of the errors falling within the range -3% to +3%. In contrasts, the ellipticity and external shear predictions show significantly larger uncertainties, with 68% of the predictions exhibiting errors between -100% and +100%. The dispersion on the Einstein radius residuals also becomes narrower and more symmetric, decreasing from $^{+0.08}_{-0.07}$ to $^{+0.05}_{-0.05}$.

For the other parameters, the predictive performance remains similar. Finally, we tested the U-Net + ResNet model on a set of 300 real Euclid observations, for which the lensing parameters were estimated through MCMC inference. The Einstein radius predictions initially collapsed toward a single value, indicating a mismatch between the simulated training data and the real observations. Retraining the U-Net + ResNet model on normalized HSC/Subaru images, rather than applying only a square-root transformation, led to improved predictions when evaluated on normalized Euclid data. This result highlights the importance of using a consistent preprocessing pipeline between the training and target datasets.

In the next section, we investigate the U-Net reconstruction in greater detail. Since the U-Net has learning to accurately extract the lens component from the input lensing images, we now examine the quality of its reconstructions.

U-Net reconstruction

Figure 11 show six samples of reconstruction produced by the U-Net. Rows 1,3 and 6 exhibit accurate reconstruction, the morphology of the arcs and ring's are well recovered which is confirmed by the low errors in the difference map. This directly helped the ResNet produce accurate predictions of the Einstein radius θ_E with a reported 0% error in table 2.

For the reconstructions in rows 2,4 and 5, the U-Net produces noticeably different morphologies for the arcs and rings compared to the expected structures. This may be explained by the fact that theses images appear noisier and contain more surrounding galaxies, which may be misinterpreted as multiple lensed images of the source. In addition, the arcs and rings are faint and more difficult to capture in the first place. This results in higher prediction errors from the ResNet, as reported in table 2.

Theses examples bring the following question : in which cases does the U-Net struggle to extract the lenses in the lensing system ? To address this question, Figure 12 shows the evolution of the Root Mean Squared Error (RMSE) of the U-Net reconstruction as a function of the Signal-to-Noise Ratio (SNR). The noise is estimated by randomly selecting one of the four image corners and computing the root mean square of the pixel values within a 20 by 20 patch. The Signal is computed as the sum of all pixels intensities in the source only image.

For the four g,r,i and z bands, the RMSE increases

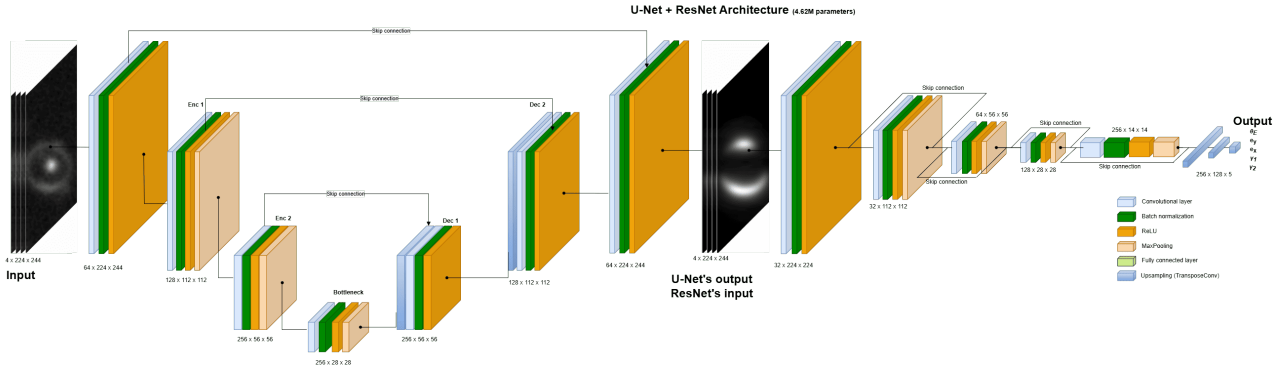


Figure 9: U-Net + ResNet model, the U-Net takes as input the lensing images and outputs the lens-only components which is then processed by the ResNet to predict the five parameters $\theta_E, e_x, e_y, \gamma_{\text{ext, 1}}, \gamma_{\text{ext, 2}}$.

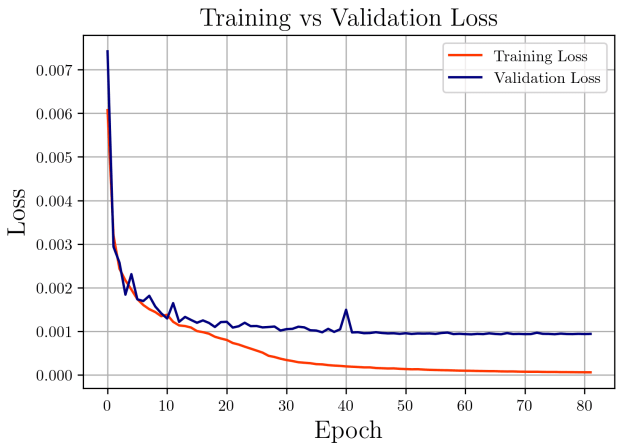


Figure 10: U-Net + ResNet's loss function during training

Image #	$\Delta\ \theta_E$ (%)	RMSE
1	0.00%	2.5788×10^{-10}
2	58.04%	9.3649×10^{-8}
3	0.00%	1.2098×10^{-8}
4	155.98%	1.5492×10^{-8}
5	63.96%	2.5166×10^{-6}
6	0.00%	3.8946×10^{-10}

Table 2: Einstein radius residuals and RMSE for different images.

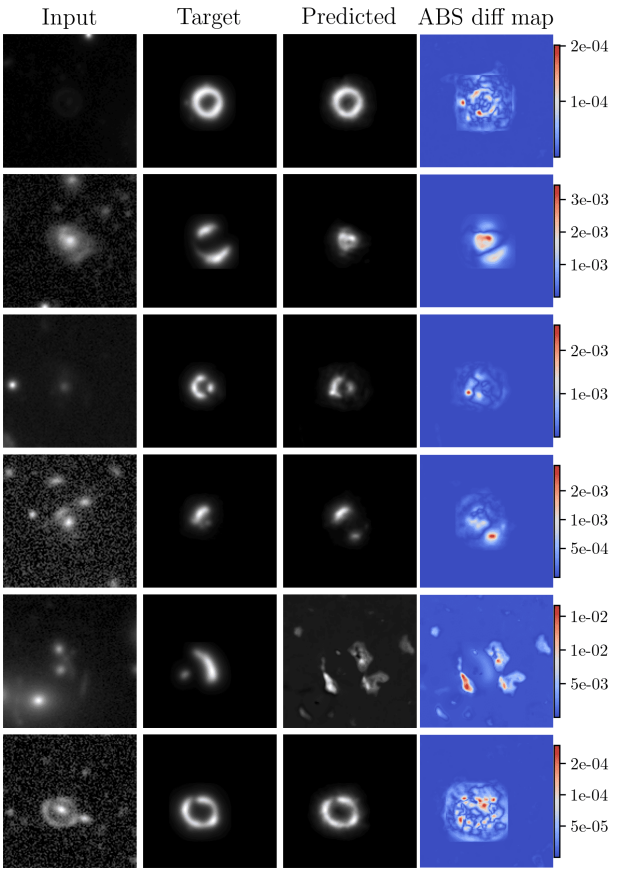


Figure 11: Six samples of U-Net's reconstruction : First column is the input image with the lensing galaxy together with the background deformed source. Second column is the background deformed source only. Third column is the attempt of the U-Net to isolate the background deformed source. Column four is the absolute difference map between the U-Net's reconstruction and the ground truth lensed source.

as the SNR decreases, as expected. Images with low SNR exhibit fainter arcs that are be more difficult to detect. This relation appears strong, as the RMSE is plotted on a logarithmic scale.

Finally, to highlight the relationship between the U-Net reconstructions and the ResNet predictions, Figure 13 shows the evolution of the residuals for the five parameters as a function of the U-Net RMSE across all four bands. A clear trend is observed for the Einstein radius, where the spreads of the residuals increases with increasing RMSE. This effect is also pronounced for the two external shear components, while it is weaker for the two ellipticity parameters.

3.3 Bayesian Neuron Network

So far, we have developed models that predicts five output values for each input sample. A natural question is whether the model can also quantify the uncertainty associated with these predictions. This capability is particularly important in scientific applications, where understanding the confidence of a prediction is often as valuable than the prediction itself. Bayesian Neural Network may address this issue.

A Bayesian neural network is a neural network that learns a distribution over its weights rather then a single best estimate. Using Bayes' theorem it infers the posterior $p(w \mid \text{data})$, which represent our beliefs about the model's weights after observing the data. This framework is particularly valuable because it provides both parameter estimates and associated uncertainty. In particular, Bayesian neural network can capture two distinct sources of uncertainties : **aleatoric uncertainty**, which arises from inherent noise in the data and **epistemic uncertainty**, which stems from limited knowledge about the model parameters.

Mathematically, a Bayesian Neural Network infers the posterior distribution : $p(w \mid \text{data})$. Since this posterior is generally intractable, it is often approximated by a simpler distribution $q(w \mid \text{data})$, such as a Gaussian distribution. To measure the quality of this distribution, we compute the Kullack-Leibler (KL) divergence between the 2 distribution :

$$\text{KL}(q \mid\mid p) = \int q(w \mid \text{data}) \log \left(\frac{q(w \mid \text{data})}{p(w \mid \text{data})} \right) dw.$$

The objective is to minimize this quantity. With a bit of calculus, one can show that this is equivalent to minimizing the negative Evidence Lower Bond (ELBO):

$$\mathcal{L} = -\mathbb{E}_{q(w|\text{data})}[\log p(\text{data} \mid w)] + \text{KL}(q(w \mid \text{data}) \mid\mid q(w))$$

The first term corresponds to the reconstruction error (e.g., mean squared error in regression tasks or cross-entropy in classification), while the second term acts as a regularization term that keeps the approximate posterior close to the prior distribution over weights.

Implementing this architecture relatively straightforward thanks to [13], which demonstrates that introducing dropout between layers, i.e, randomly removing connections during training, is sufficient to approximate variational inference. The KL term in the objective function can be approximated by a simple regularization term $|| w ||^2$, which corresponds to a L_2 weight penalty term.

This technique allows one to estimate the **epistemic uncertainty** by passing the same input through the model multiple times and computing the variance (or standard deviation) of the outputs. To estimate the **aleatoric uncertainty**, one must modify the reconstruction function, replacing the mean squared error with a Gaussian log-likelihood :

$$NLL = -\log p(y \mid x, w) = \frac{1}{2\sigma^2} ||y - \hat{y}||^2 + \frac{1}{2} \log \sigma^2 + C. \tag{3}$$

Practical implementation of the Bayesian Neural Network

The first implementation of the Bayesian Neural Network consisted of adding dropout layers between every layer of the ResNet model, and doubling the output dimension to obtain five means and five variances, or log-variances. Predicting the log-variance improves numerical stability and guarantees a positive variance after exponentiation. The accuracy of the five means predicted were similar to those obtained with the classical ResNet, but the aleatoric uncertainty was constant for every input and for each of the five predicted parameters.

This can be explained by the fact that the model is already quite accurate, so the term $||y - \hat{y}||^2$ in eq. 3 is already small. As a result, the model quickly learns that it can further reduce the loss by pushing $\log \sigma^2$ towards very small values, effectively collapsing the predicted variance and reaching a degenerate solution.

To prevent this behavior, we introduce a regularization term controlled by a parameter (β) into the Gaussian log-likelihood. This additional term penalizes excessively large value of the predicted log-variance, preventing the network from artificially increasing its uncertainty to reduce the reconstruction error. In practice, we set (β) to small values typically (0.01) or (0.005). The modified Gaussian negative log-likelihood is therefore defined as :

$$NLL = \frac{1}{2\sigma^2} ||y - \hat{y}||^2 + \frac{1}{2} \log \sigma^2 + \beta \ (\log \sigma^2)^2 \tag{4}$$

Furthermore, We modified the regression head of the ResNet, by splitting the regression head into two branches : one dedicated to estimating the five mean values and the other to estimating the five corresponding log-variances. This separation is essential because the mean and variance represent fundamentally different quantities. The log-variance branch was further expanded with additional layers, increasing its capacity to learn a more complex uncertainty representation

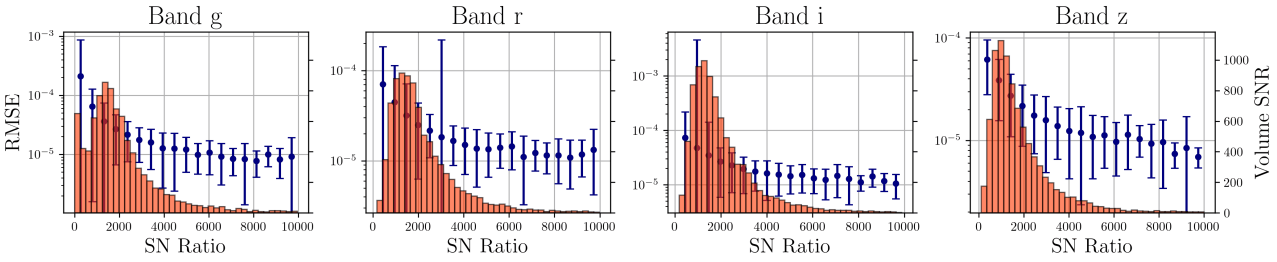


Figure 12: Root Mean Squared Error (RMSE) as a function of the Signal to Noise Ratio (SNR) for images of the test set for each of the four g,r,i and z bands.

and reducing the tendency of the model to collapse towards fixed variance values.

Training and Results

To train this new model we retained the same hyperparameter as those used for the U-Net + ResNet architecture. In addition, dropout layers were introduced between each layers of the regression head, with a dropout rate of (0.2) Following the approach proposed by Gal and Ghahramani [13]. The model was trained for up to 100 epochs, with early stopping triggered after 65 epochs.

As shown in Figure 14, the training process remained stable and exhibited no visible signs of overfitting. Although the loss function takes negative values in this configuration, it still converges properly, indicating successful optimization of the model.

The results shown in Figure 15 indicate that the predictive performance on the test set is largely preserved for the Einstein radius and the two ellipticity components. In contrasts, the predictions for the two shear components deteriorate significantly. The model tends to predict values that are clustered around the mean of the distribution, failing to capture the true variability of the shear parameters. This suggests that, by increasing the number of target parameters to be predicted simultaneously, the model loses its ability to extract and retain the information necessary for accurate shear estimation.

One possible approach to mitigate this degradation is to assign different weights to the five target parameters in the loss function, placing greater emphasis on the recovery of the shear components. This strategy was previously tested with the ResNet architecture. However, the results showed a degradation in the prediction accuracy of the Einstein radius and ellipticity components, while providing little to no improvement in the estimation of the shear parameters. Consequently, parameter weighting does not appear to offer an effective solution to the loss of the shear information observed in the multi-parameter prediction setting.

Uncertainty calibration

Since the five uncertainties estimates produced by the model are learned in an unsupervised manner, their true values are not available and therefore cannot be assessed directly. As a result, these uncertainties must be calibrated to ensure that they accurately reflect the model's predictive confidence. Ideally, for a well-calibrated Gaussian uncertainty estimate, approximately 68% of the predictions should lie within one predicted standard deviation of the true value, i.e., within the interval defined by the predictive value ± its associated uncertainty. Achieving this coverage would indicate that the predicted uncertainties are consistent with their intended probabilistic interpretation.

We adopt the same uncertainty calibration procedure as that proposed in the LEMON Framework [14]. After training the model, the predicted uncertainties are evaluated on a small subset of the testing set and are interpreted as the aleatoric uncertainty. To estimate the epistemic uncertainty, we perform 100 forward passes through the network, generating a distribution of the predicted mean values. The standard deviation of these predicted means is then used as an estimate of the epistemic uncertainty. Finally, the total predictive uncertainty is obtained by combining the aleatoric and epistemic contributions through their quadratic sum. This resulting uncertainty estimate is then used in the subsequent calibration procedure.

For each parameter, we compute the corresponding z-scores using the predicted uncertainties and compare their empirical cumulative distribution function (CDF) with the theoretical CDF of a standard Gaussian distribution (see Fig 16). To quantify the discrepancy between the two distribution, the empirical CDF is fitted using the interpolation function.

$$\beta(x, a, b, c) = \frac{1}{1 + e^{c} \frac{x^{a}}{(1-x)^{b}}} \tag{5}$$

.

whose parameters are determined through a fitting procedure. Once the interpolation function has been obtained, an optimization problem is solved to determine the calibration coefficient (β) that best maps the interpolated curve onto the diagonal line corresponding to perfect calibration. In other words, (β) rescales the predicted uncertainties so that the empirical coverage matches the expected Gaussian coverage (see Fig 16 for before and after the calibration procedure).

This calibration procedure is applied independently to each of the five predicted parameters : the Einstein

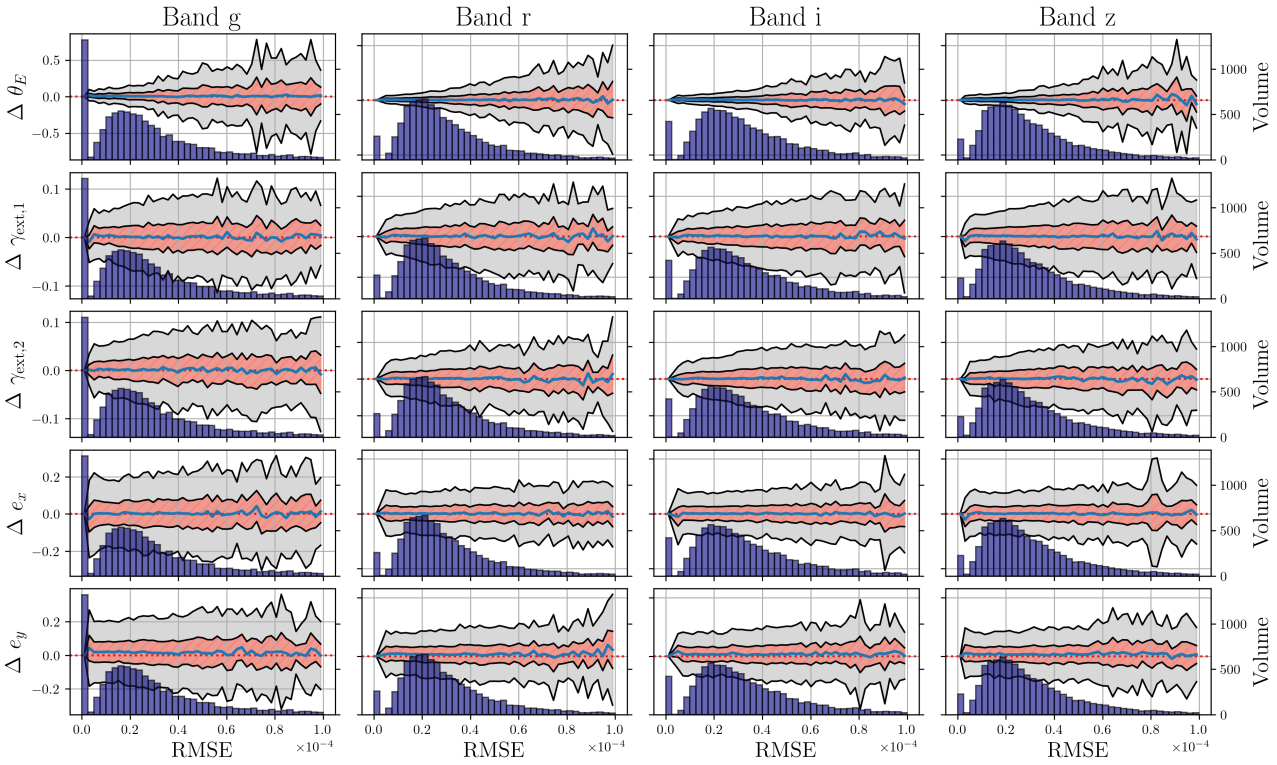


Figure 13: ResNet's residuals as a function of the Root Mean Squared Error (RMSE) of the U-Net reconstruction. The solid blue line represents the median trend, while the salmon and gray bands indicates the 1σ and 3σ confidence intervals, respectively.

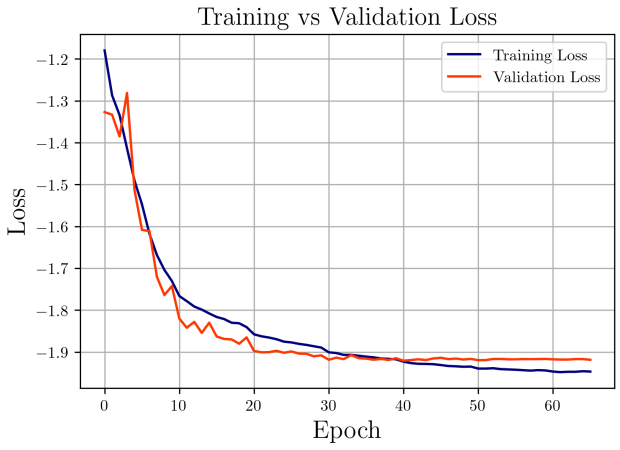


Figure 14: Bayesian ResNet's loss function during training

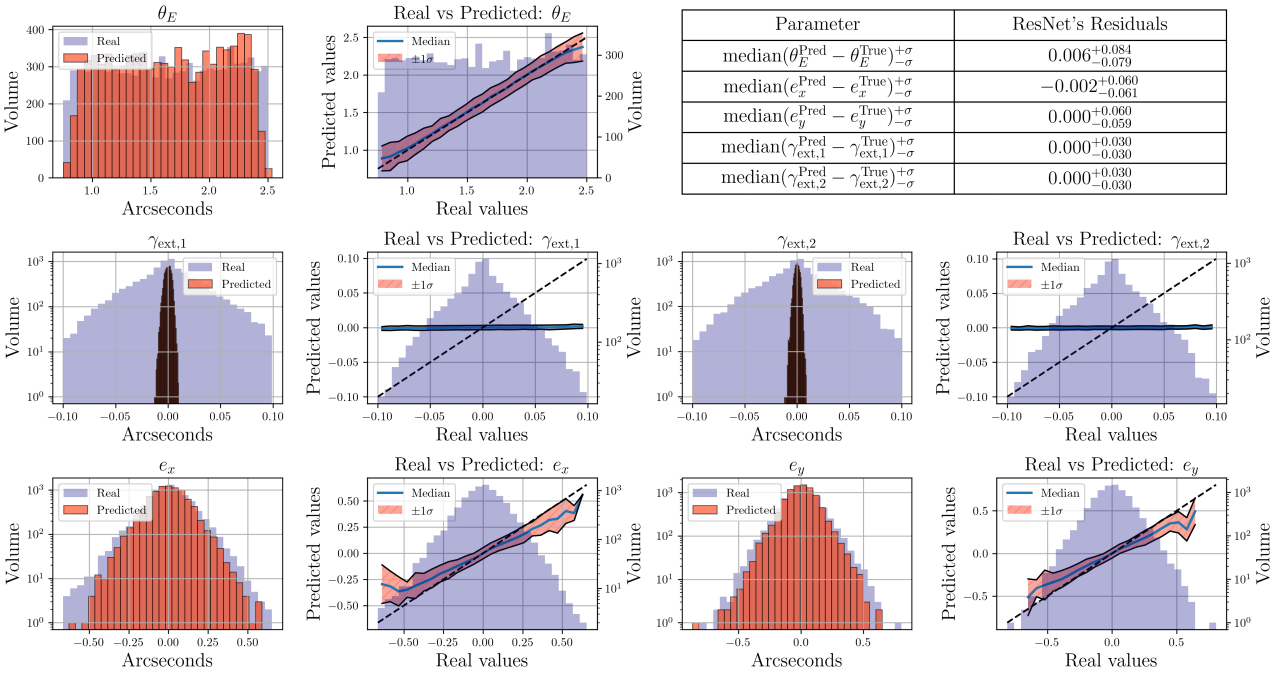
radius, the two ellipticity components, and the two shear components. Consequently, a distinct calibration coefficient is obtained for each parameter, yielding individually calibrated uncertainty estimates.

Once the uncertainties have been calibrated, we can visualize the predicted mean values along with their associated total uncertainties. Figure 17 presents the predictions and calibrated uncertainties for the five target parameters.

For the Einstein radius, we observe that most predicted uncertainty intervals encompass the true values, represented by the diagonal line. A fraction of the predictions fall outside their corresponding uncertainty intervals, which is consistent with the expected statistical interpretation of the calibration. Indeed, for a correctly calibrated Gaussian uncertainty, around 32% of the values are expected to fall outside this interval.

For the shear and ellipticity components, the predicted uncertainty intervals are significantly larger, highlighting the greater difficulty faced by the model in accurately reconstructing these parameters. The increased uncertainty reflects the lower predictive confidence of the model and indicates that these quantities are more challenging to infer from the input data compared to the Einstein radius.

The σ coverage for the five parameters shown in the top-right table of Fig. 17 is very close to the theoretical value of 68%, indicating that the calibration procedure was successful. In Figure 16, we observe that, before calibration, the predicted uncertainties are



Parameter	ResNet's Residuals
$\text{median}(\theta_E^{\rm Pred} - \theta_E^{\rm True})^{+\sigma}_{-\sigma}$	$0.006^{+0.084}_{-0.079}$
$\text{median}(e_x^{\rm Pred} - e_x^{\rm True})^{+\sigma}_{-\sigma}$	$-0.002^{+0.060}_{-0.061}$
$\text{median}(e_y^{\rm Pred} - e_y^{\rm True})^{+\sigma}_{-\sigma}$	$0.000^{+0.060}_{-0.059}$
$\text{median}(\gamma_{\rm ext,1}^{\rm Pred} - \gamma_{\rm ext,1}^{\rm True})^{+\sigma}_{-\sigma}$	$0.000^{+0.030}_{-0.030}$
$\text{median}(\gamma_{\rm ext,2}^{\rm Pred} - \gamma_{\rm ext,2}^{\rm True})^{+\sigma}_{-\sigma}$	$0.000^{+0.030}_{-0.030}$

Figure 15: Performance of the Bayesian ResNet on the test set (approx. 8000 samples). For each of the five parameters, two panels are shown. The left panel compares the distribution of the predicted values the ground truth. The right panel shows the prediction values as function of the ground-truth values. The median prediction (solid blue line) and the corresponding 1 σ confidence band are overlaid. The table in the upper-right corner summarizes the median residuals and their corresponding 1 σ intervals for all five parameters.

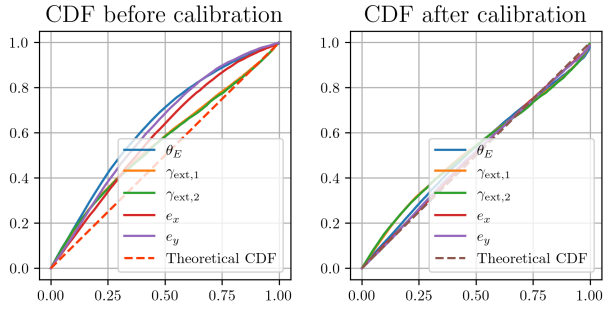


Figure 16: Uncertainties calibration for the five parameters.

overestimated, as all curves lie above the diagonal line. Consequently, the recalibration factor is then smaller than one (see column β of the top-right table), reducing the predicted uncertainties and bringing them into better agreement with the expected confidence levels.

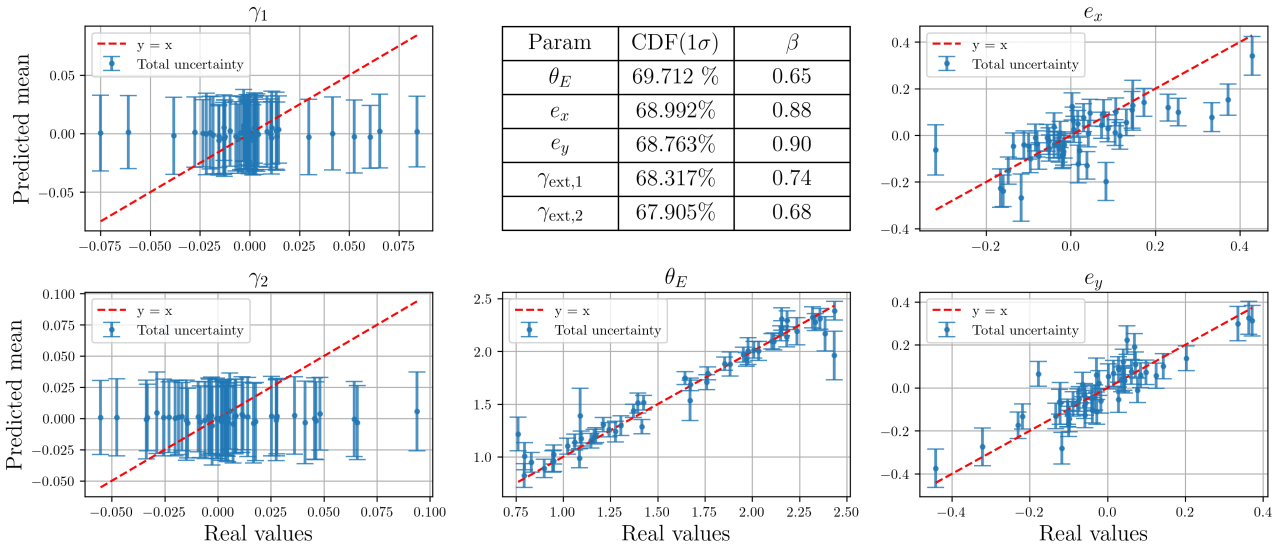
Finally, to visualize those uncertainties directly on the images, Figure 18 shows the predicted Einstein rings overlaid on the simulated images. The green ring represents the true radius. In some examples, it is not visible because, it overlaps with the prediction or falls on the limit of the uncertainty region, indicating strong precision in the model's predictions. The uncertainty appears narrow for most predictions, suggesting that the model is confident in its estimates. However, for the example in the top-right, the uncertainty region is larger, meaning lower confidence from the model, potentially due to the information contained in the image or uncertainty arising from the model's parameters themselves.

U-Net + ResNet on Euclid simulation

To further evaluate the performance of our U-Net + ResNet model, we tested it on a new dataset composed entirely of simulated Euclid observations. In this dataset, both the source galaxies and the lensing galaxies are generated exclusively from Euclid data. Examples of these simulated lenses are shown in Figure 19. Unlike the previous dataset, which included four photometric bands (g, r, i, z), this dataset contains only a single band. The total dataset comprises approximately 11,000 samples.

The training time of the model is relatively short, remaining under two hours due to the reduced size of the dataset. The training loss and validation loss converges rapidly to a minimum and demonstrating stable training behavior with no significant sign of overfitting (see Fig. 20).

Figure 22 presents the prediction performance of the U-Net + ResNet model for the five lensing parameters on the Euclid dataset. The results are consistent with those obtained on the Subaru/HSC dataset, despite the significantly higher signal-to-noise ratio (SNR) of the Euclid images (see Fig. 21). The lensed arcs and Einstein rings produced by the background sources are substantially brighter than those in the Subaru/HSC dataset. However, the increase in image resolution and



Param	CDF(1σ)	β
θ_E	69.712 %	0.65
e_x	68.992%	0.88
e_y	68.763%	0.90
$\gamma_{\text{ext},1}$	68.317%	0.74
$\gamma_{\text{ext},2}$	67.905%	0.68

Figure 17: Prediction versus ground-truth values for each of the five parameters, including the total uncertainty for 50 samples. The table in the top-right corner shows the empirical coverage of the predicted uncertainties across the entire test set (8,000 samples) along with their respective recalibration factor β.

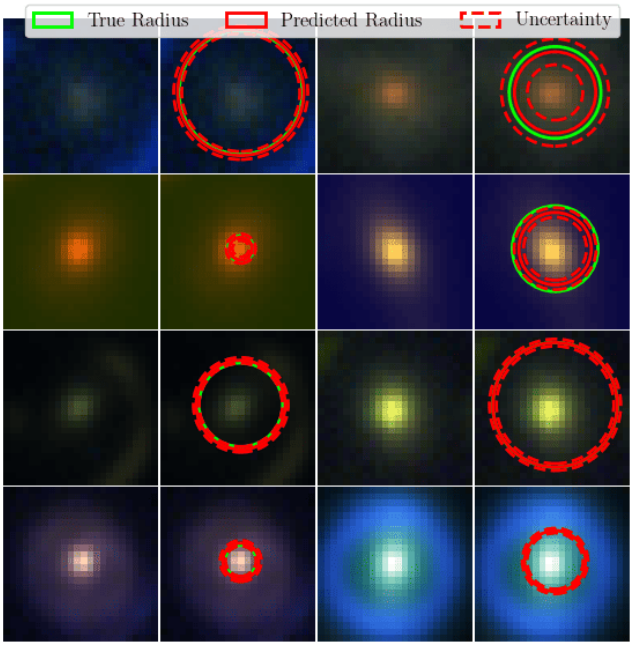


Figure 18: Eight examples of predictions made by the Bayesian ResNet for the Einstein Radius and their associated uncertainty. For each of the eight images, the left panel shows the original image, while the right panel displays the predicted Einstein radius (red), the true radius (green), and the corresponding uncertainty region (dashed line) overlaid on the image.

SNR does not lead to a significant improvement in parameter recovery. In particular, the predictions of the external shear and ellipticity remain unchanged, whereas the Einstein radius continues to be accurately recovered. We investigated the impact of reducing the input dimensionality by switching from four input images to a single image. To account for the reduced number of input channels, we tested two modifications of the ResNet architecture: reducing the number of channels in the ResNet layers by a factor of 4 and reducing the dimensionality of the regression head by a factor of 4. Neither modification resulted in any significant improvement in performance. We also tested removing images with extremely high signal-to-noise ratios (SNR > 7000) from the dataset, but this filtering step did not lead to any improvement either.

4 Conclusion

The implemented deep learning models can infer lensing potential parameters within milliseconds. The training process is relatively fast, requiring approximately one hour for the smaller model (ResNet) and less than ten hours when adding the U-Net. Among the predicted parameters, the Einstein radius is recovered with high accuracy, with the U-Net + ResNet achieving the best performance (68% of the predictions within $\pm 3\%$). However, all models struggle to infer the external shear. Switching to high-resolution Euclid data did not improve the model's predictive performance for the external shear parameter. The remaining test to be conducted is to train the model on a dataset that is evenly distributed across the external shear parameter space, in order to assess whether the current performance is limited by the parameter distribution of the training data.

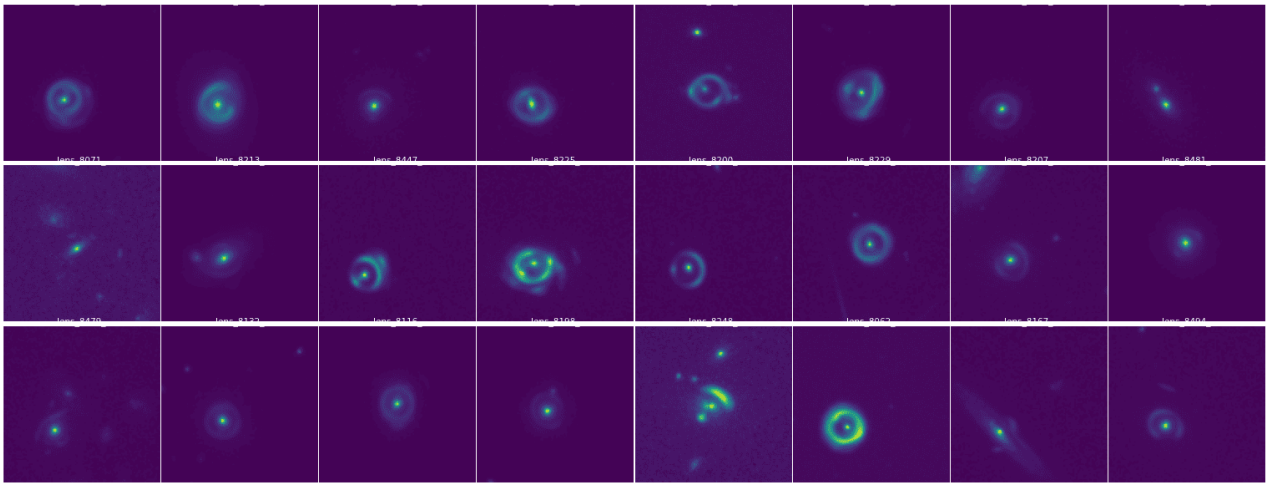


Figure 19: 24 examples of the dataset simulated with Euclid images

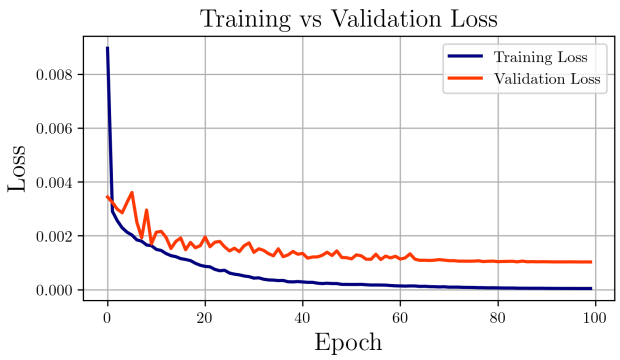


Figure 20: U-Net + ResNet training and validation loss on the dataset generated from Euclid images.

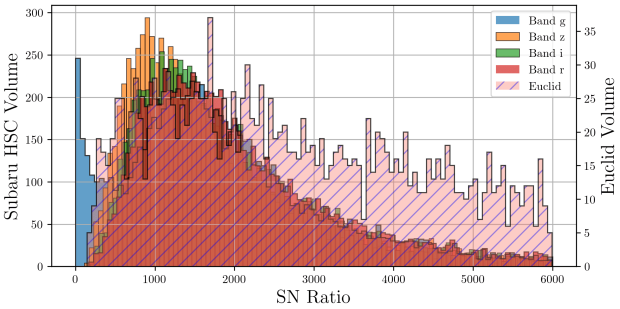


Figure 21: Signal to noise ration (SNR) of the Subaru/HSC (left) and the Euclid dataset (right).

The U-Net effectively isolates lensing systems, making it a useful preprocessing step for subsequent MCMC inference. The Bayesian Neural Network is straightforward to implement and provides meaningful uncertainty estimates, provided that these uncertainties are properly calibrated. This work provides a baseline model for future Euclid analysis and for the identification of lensed Fast Radio Burst (FRBs).

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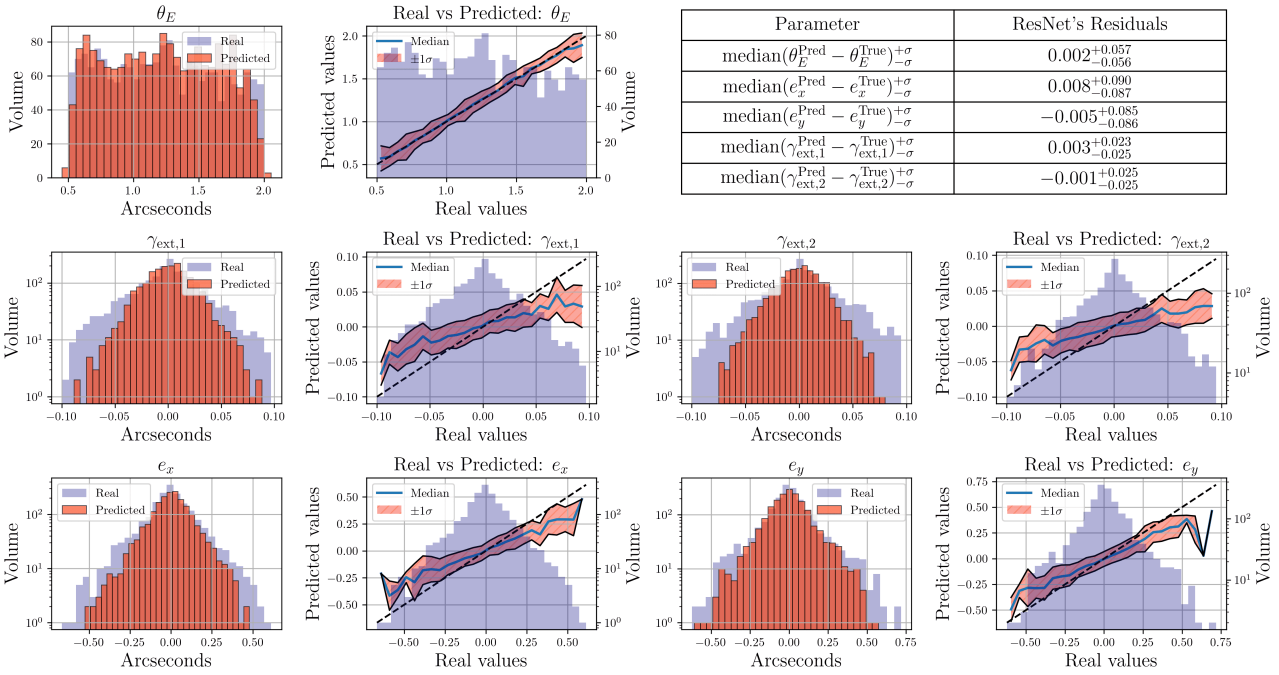
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Parameter	ResNet's Residuals
$\mathrm{median}(\theta_E^{\mathrm{Pred}} - \theta_E^{\mathrm{True}})^{+\sigma}_{-\sigma}$	$0.002^{+0.057}_{-0.056}$
$\mathrm{median}(e_x^{\mathrm{Pred}} - e_x^{\mathrm{True}})^{+\sigma}_{-\sigma}$	$0.008^{+0.090}_{-0.087}$
$\mathrm{median}(e_y^{\mathrm{Pred}} - e_y^{\mathrm{True}})^{+\sigma}_{-\sigma}$	$-0.005^{+0.085}_{-0.086}$
$\mathrm{median}(\gamma_{\mathrm{ext},1}^{\mathrm{Pred}} - \gamma_{\mathrm{ext},1}^{\mathrm{True}})^{+\sigma}_{-\sigma}$	$0.003^{+0.023}_{-0.025}$
$\mathrm{median}(\gamma_{\mathrm{ext},2}^{\mathrm{Pred}} - \gamma_{\mathrm{ext},2}^{\mathrm{True}})^{+\sigma}_{-\sigma}$	$-0.001^{+0.025}_{-0.025}$

Figure 22: Performance of the U-Net + ResNet on the Euclid test set (approx. 2000 samples). For each of the five parameters, two panels are shown. The left panel compares the distribution of the predicted values the ground truth. The right panel shows the prediction values as function of the ground-truth values. The median prediction (solid blue line) and the corresponding 1 σ confidence band are overlaid. The table in the upper-right corner summarizes the median residuals and their corresponding 1 σ intervals for all five parameters.

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A Appendix

A.1 Vision Transformer

A Vision Transformer (ViT) is a deep learning architecture introduced in 2020 that applies the Transformer mechanism, originally developed for natural language processing, to image analysis. Transformers are highly effective at capturing long-range dependencies within data, which is why similar architectures are used in models such as ChatGPT and Claude for processing text.

A Vision Transformer first divides an input image into small fixed-sized patches, which are treated as visual tokens. Each patch is then converted into a numerical Vector through an embedding layer, projecting the image information into a high dimensional feature space. In this space, similar visual features tends to be represented by nearby vectors.

$$Z_i = EX_i + E_{\text{loc}} \tag{6}$$

where X_i corresponds to path i of the image X, E is the learned embedding matrix, and E_{loc} encodes the positional information of patch i within the image.

The model then uses self-attention blocks to determine how each token relates to all the others by computing three quantities : the query (Q), Key (K), and Value (V) representations,

$$Q = W_Q Z, \quad K = W_K Z, \quad V = W_V Z \tag{7}$$

where W_Q, W_K and W_V are learned projection matrices. The attention mechanism computes the similarity between token representations and assigns different weights depending on how much information each patch provides about the others :

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{8}$$

Here QK^T measures the similarity between all pairs of tokens, while the factor $\sqrt{d_k}$ normalizes the dot products, preventing them from becoming excessively large as the dimensionality d_k increases. The Softmax functions converts these similarity scores distribution, such that each row of the attention matrix A sums to one. The final attention output is then obtained by multiplying A with V. This allows the model to capture relationships between different region of the image, rather than relying only on local features as traditional convolution neural networks do.

A Multi Layer Perceptron (MLP) then processes each token representation to learn non-linear features :

$$\text{MLP}(AV) = W_2\sigma(W_1 AV + b_1) + b_2 \tag{9}$$

where W_1 and W_2 are the learned weights matrices, b_1 and b_2 are bias terms, and σ is a non-linear activation function, typically GELU.

A transformer is composed of several stacked transformer blocks where the same sequence of operations is repeated: self-attention, a multilayer perceptron (MLP), residual connections, and layer normalization. At the end of the network, a pooled representation of the tokens is extracted and passed to a prediction head :

$$Z = \text{Transformer}(X) \tag{10}$$

$$\hat{Y} = \text{FC}_{\text{head}}(Z) \tag{11}$$

where Fc_{head} is a fully connected regression head that maps the learned representation Z to the predicted parameters $\hat{Y}$.

A.2 Practical implementation of the Vision Transformer

Vision Transformer are typically large models that require substantial datasets for training. However, many pretrained models are available and can be adapted for specific applications. In this work, we use the pretrained DINOv2 ViT-S/14 model developed by Meta, which consists of 14 Transformer blocks and contains 22.15 million parameters. The model was pretrained on a large-scale dataset containing approximately 142 million images from the LVD-142M dataset (Large-scale Vision Dataset). The dataset contains a wide range of visual concepts such as objects, scenes, animals and textures.

We attach a regression head with dimensions 382 by 256 by 5 to adapt the DINOv2 transformer to our study. The model must then be fine-tuned on our gravitational lensing dataset. An important question is whether all 14 Transformer blocks need to be retrained, or whether only a subset of the blocks should be updated. To investigate this, we will draw inspiration from the approach presented by S. H. Vincken et al. [17].

The idea is to evaluate the contribution of each transformer block by computing the MSE after progressively including additional blocks, starting from the first Transformer block. By plotting the MSE as a function of the block index, the largest decrease will indicate the optimal point from which Transformer blocks should be fine-tuned, while keeping earlier blocks frozen. Figure 23 presents the results of this analysis. As observed, the MSE decreases sharply after block 8, suggesting that block 1 to 8 could be frozen while retraining block 8 to 14 with an appropriately selected learning rate.

A.3 Other test performed

❖ **Tests conducted on the ResNet**

- *Images transformations :* We investigate the impact of different image preprocessing techniques, including standard normalization, an arcsinh transformation, and no transformation. The best performance is obtained by shifting the image negative values to zero and applying a square-root transformation.
- *Learning rate scheduler :* We tested several learning rate scheduling strategies to reduced the learning rate during training, including

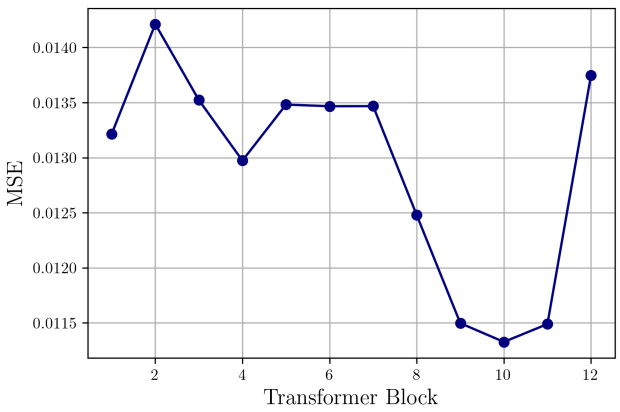


Figure 23: Evolution of the mean squared error (MSE) as a function of the number of transformer blocks in the Vision Transformer model.

ReduceLROnPlateau, a step scheduler dividing the learning rate by a factor of γ every x epochs, and cosine annealing with warmup. Among these approaches, the best performance was achieved using a cosine annealing scheduler.

- *Global average and flattening :* At the output of the last residual block of the ResNet, the feature map consists of 256 channels with a spatial size of 17 by 17. We tested different strategies to aggregate these features, including global average pooling, average pooling with kernel sizes of 4 by 4, and 8 by 8, as well as flattening. The flattening approach was not feasible, as it significantly increased the number of parameters and exceeded the available GPU memory capacity. Overall, increasing the number of parameters did not lead to significant performance improvements. The best results were obtained using global average pooling, which provides an efficient feature representation while maintaining good predictive performance.

❖ **Tests conducted on the U-Net**

- *U-Net depth :* We investigated the impact of the U-Net depth on reconstruction accuracy. Reducing the depth from 4 to 3 levels improved the reconstruction performance. We also tested a U-Net++ architecture, but despite its increased complexity and computational cost, it did not provide any significant improvement over the standard U-Net model.

❖ **Tests conducted on the U-Net + ResNet**

- *ResNet connection :* We connected the ResNet to the bottleneck of the U-Net, but this modification did not lead to any improvement in performance.
- *Weighted loss functions :* For the U-Net + ResNet architecture, we tested combining two loss functions with different weighting factors. The best results were obtained when the weighting factor balanced the two loss contributions, bringing them to the same order of magnitude.
- *Adding additional context :* We investigated whether providing the ResNet with additional contextual information could improve the parameter prediction. Instead of using only the lensing region extracted by the U-Net, we provided the ResNet with both the U-Net predicted lens-only image and the residual information obtained by subtracting this prediction from the original input image. This approach was intended to preserve information from the surrounding environment while maintaining the isolated lensing features. However, this additional context did not lead to any significant improvement in performance.